

UX flows & interaction patterns — the end-to-end behaviour of every HelixVPN surface

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Master technical specification — Volume 10 (Design System), nano-detail deep-dive. This document **owns** HelixVPN's **end-to-end UX flows and the universal interaction patterns** that bind the three apps (Client, Console, Connector) across eight platforms and four form factors into one coherent product. It is the **behavioural** half of the design system: where [screens-client.md] / [screens-console.md] / [screens-connector.md] own *each screen* and [component-library.md] owns *each component*, this doc owns the **journeys between them** — the step-by-step sequences, the per-app information architecture + navigation model, the universal interaction grammar (gestures, keyboard, TV-remote D-pad, long-press, pull-to-refresh), the reusable **loading / empty / error / success** state contract, the **motion choreography**, the full **accessibility** spec (WCAG 2.1 AA target), and the **i18n / RTL / localization** patterns.

The product's emotional centre of gravity is one question — "*am I protected right now?*" [04_CLIENT §7]. Every flow below is designed so that question is answered instantly and unambiguously, and so the **loading / connecting** state is never mistaken for the **connected** state (§11.4.107 — a spinner is **not** proof of connection; the connected state is reached only on an actual Connected{...} FFI event, never on a timer or an optimistic guess).

SPEC-ONLY. It describes *what each flow is and how each interaction behaves* — not the shipping Flutter navigator, gesture recognisers, or focus-traversal code. Screen layouts are owned by the three screens-* docs; component internals by [component-library.md]; colours by [color-system.md]; token scales (spacing / motion / breakpoints) by [design-tokens.md]; the type/icon/motion catalog by [typography-iconography-motion.md].

Boundary with sibling docs. Owns: the cross-screen **journeys**, the per-app **navigation model**, the **interaction grammar**, the **state-pattern contract**, **motion choreography**, the **a11y flow**, and **i18n/RTL**. **Consumes:** the **7-variant ffi::TunnelStatus** state machine + its Dart-facing state diagram [FFI §3.2, §3.4]; the connection-state palette [COLOR §3]; the motion tokens [DT §6.4]; the screen inventories + nav maps of the three screens-* docs; the components each flow drives [component-library.md].

Evidence base. [04_CLIENT §N] = final/03-client-core-and-ui.md; [FFI §N] = final/v04-client/ffi-surface.md; [SC §N] = final/v10-design/screens-client.md; [SK §N] = final/v10-design/screens-console.md; [SN §N] = final/v10-design/screens-connector.md; [CL §N] = final/v10-design/component-library.md; [COLOR §N] = final/v10-design/color-system.md; [DT §N] = final/v10-design/design-tokens.md; [CP §N] = final/02-control-plane.md; [POL §N]/[REG §N]/[IPAM §N]/[COORD §N] = the matching final/v03-control-plane/svc-*.md. Claims not grounded in the evidence base or in this document's own original UX design choices are tagged UNVERIFIED per constitution §11.4.6 — never fabricated. The flow and interaction designs are **original HelixVPN UX design work** (modelled on well-established WAI-ARIA Authoring-Practices, Material/Apple-HIG, and TV-leanback focus patterns, cited in Sources).

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0. Scope & the principles that shape every flow

This document specifies the **dynamic** behaviour of HelixVPN: the ordered sequences a user moves through, and the grammar of interaction those sequences share. Six principles govern every flow:

1. **The status stream is the single source of truth.** Every connection-bearing screen is a **pure function** of the `ffi::TunnelStatus` stream [FFI §3.4]; no flow ever paints "Protected" off a timer, an optimistic local guess, or a button-press it merely *initiated*. Only an actual `Connected{...}` event paints green (§11.4.107, [04_CLIENT §8.2]).
2. **Danger overrides intent.** A `Danger{kind}` event interrupts whatever the user is doing and pushes the red surface to the top of the stack [FFI §3.3, SC §11]. No flow can suppress it; every flow yields to it.
3. **No dead ends.** Every error, empty, denied, or blocked state offers a next action (retry, re-explain, deep-link, alternative path). A flow never strands the user (§11.4.66 — closed-set choices use the interactive question mechanism, never a silent wait).
4. **Honesty over optimism (§11.4.6).** A flow shows *cached + stale-labelled* data rather than a fake-fresh value; a "loading" spinner is visually + semantically distinct from a "connected" success; a partial result is shown as partial, never rounded up to done.
5. **One decision per step.** Onboarding and destructive flows present a single decision per screen (the §11.4.162 no-overlap covenant applied to time, not just space) so the path is unambiguous on every form factor.

6. **Reachable on every input model.** Each flow is fully operable by touch, keyboard, screen reader, and TV-remote D-pad — the canonical path is never gesture-only or hover-only (§13).

Honest boundary (§11.4.6). This doc owns *flow + interaction behaviour*. The exact router config is deferred to a Volume-10 navigation refinement pass (see MASTER_INDEX; not yet authored); the gesture-recogniser thresholds and focus-traversal matrices are pinned by [platform-adaptation.md]; where a concrete numeric threshold or per-platform focus order is named without a grounding citation it is tagged UNVERIFIED.

1. Information architecture & navigation model (per app, per form factor)

1.1 The navigation-affordance ladder

All three apps share one responsive law: **the navigation affordance is chosen by width size-class, never by Platform.isX** [04_CLIENT §7.3, SC §0, SK §1, SN §1], so a resized desktop window degrades to a phone layout and Web/desktop "just work". The shared ladder:

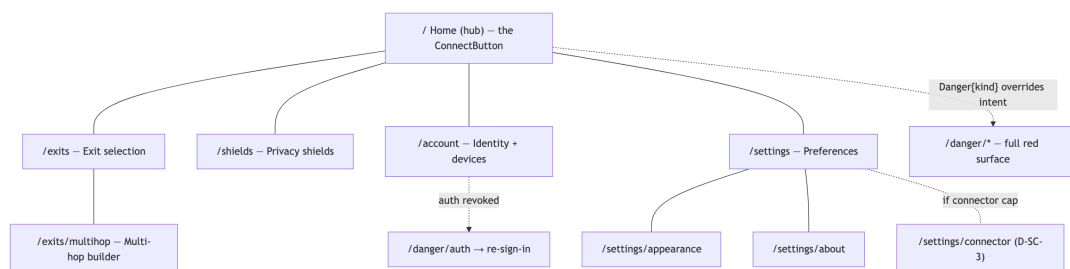
Size class	Width [DT §6.6]	Nav affordance	Body	Archetype
compact	< 600 px	BottomNavigationBar	single pane	phone (portrait)
medium	600–1024 px	NavigationRail (icon-only)	master / detail	tablet / landscape phone
expanded	> 1024 px	extended NavigationRail (icon + label)	multi-pane	desktop
tv	≥ 1440 px + leanback input	extended rail, enlarged focus targets, D-pad traversal	multi-pane	Android TV / leanback

Drawer is **not** a primary HelixVPN nav affordance: the destination count per app is small (Client 5, Console 10, Connector 3), so a bottom-nav (compact) or rail (medium+) holds them all without a hamburger-hidden menu. A drawer is used only for **secondary overflow** where a screen genuinely needs it (Console's "More" sheet on compact for destinations beyond the bottom-nav's 5 [SK §1]). This is **D-UX-1**.

TV-leanback is a size class + an input model, not a separate platform tree [SC §0]. It renders the expanded layout with focus targets enlarged to a ~64 px hit area, a visible focus ring (border.focus [COLOR §4.5]) on every focusable, and a per-screen declared D-pad traversal order (§10.3). The Console and Connector also render on TV via the same ladder, but the **Client** is the only flavor whose hero (the ConnectButton) is designed as the default-focused TV control [SC §5].

1.2 Client IA

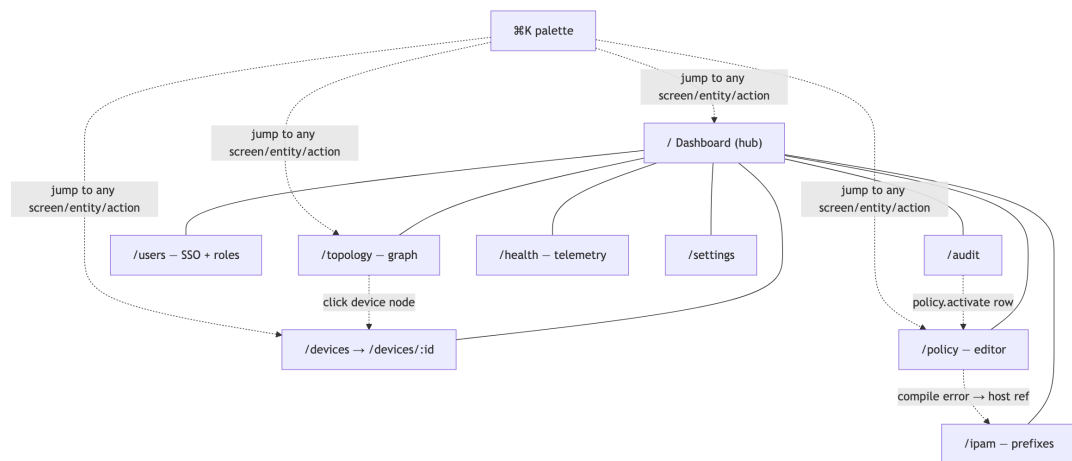
The Client is a **hub-and-spoke from Home** [SC §2]. Home (/) is the root of the post-onboarding AdaptiveScaffold; Exits, Shields, Account, Settings are its peer destinations.



- **compact:** bottom-nav with 5 destinations (Home · Exits · Shields · Account · Settings); Home shows the full-width ConnectButton [SC §3].
- **medium:** icon-only rail; Exits and Settings become master/detail (list left, selected detail right) without leaving the destination.
- **expanded:** extended rail; Home gains a right pane (live status detail), Exits gains a wide detail pane.
- **tv:** extended rail; ConnectButton is the default-focused control; D-pad down reaches the exit summary then the shields row [SC §5].

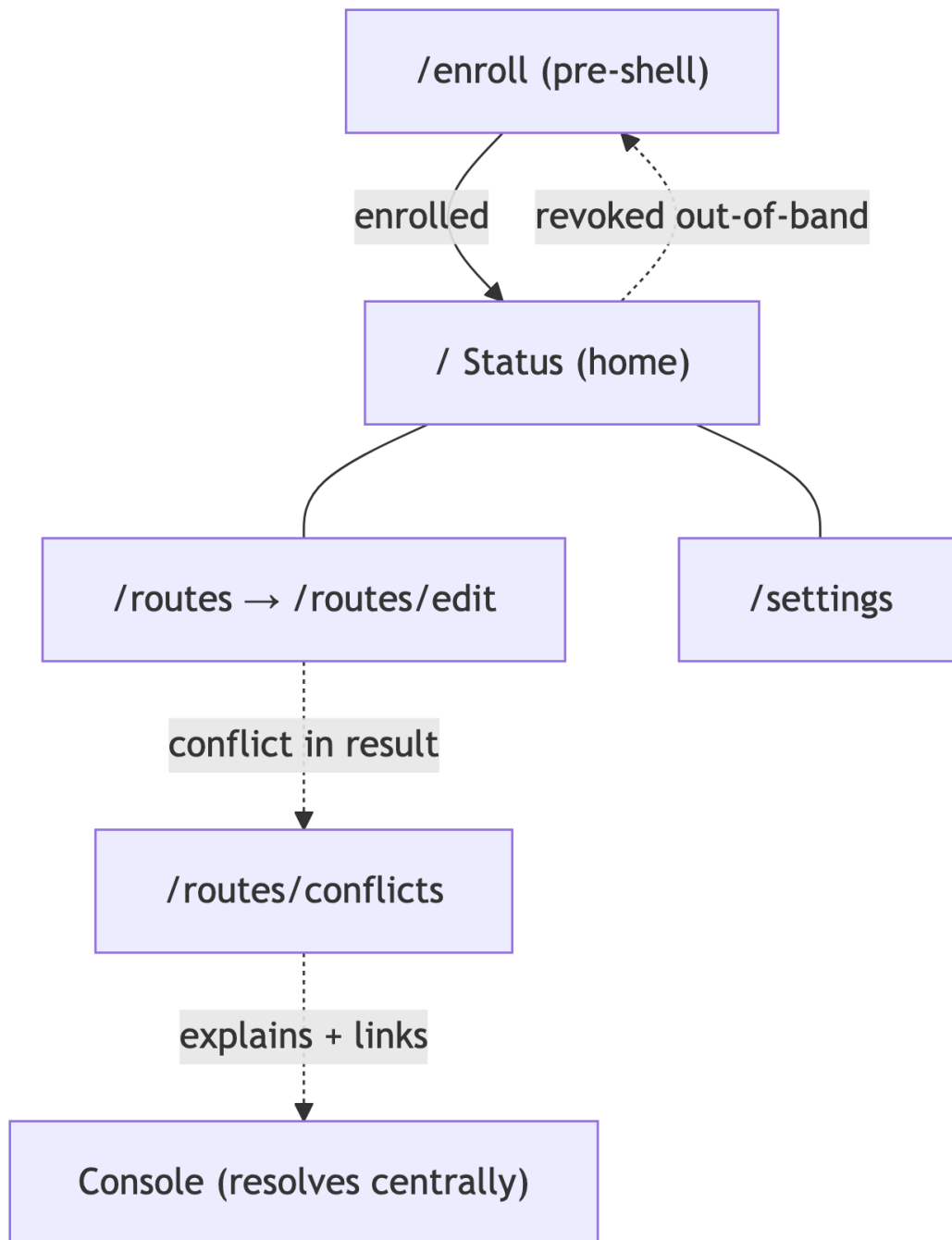
1.3 Console IA

The Console is a **hub-and-spoke from the Dashboard** with **cross-links that follow the data** [SK §2] (a policy compile error naming an un-advertised host CIDR links to IPAM; a topology connector node links to its prefixes; an audit policy.activate row links to that policy version). It adds a **⌘K command palette** [SK §1] as a keyboard-first jump-to-anything affordance orthogonal to the rail, and a **live-events rail** (expanded only) folding the WS/SSE stream.



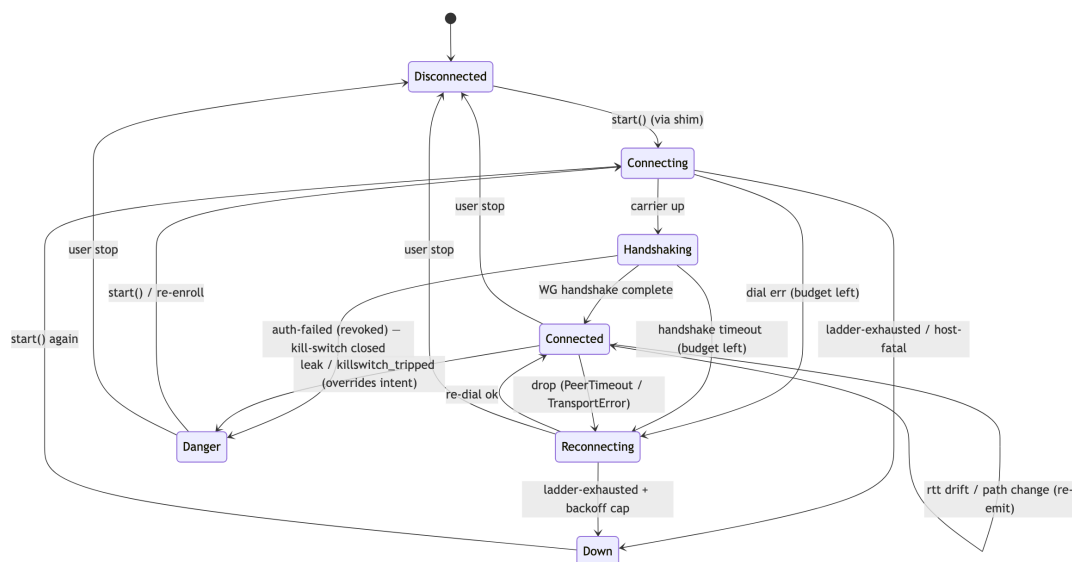
1.4 Connector IA

The Connector is a **near-linear flow** [SN §2] with a deliberately narrow IA: get enrolled, prove healthy, declare subnets, surface conflicts. Three primary destinations only (Status · Routes · Settings); Enroll precedes the shell.



2. The state-machine spine every flow rides

Every connection-bearing flow (Client Home, Connector Status) switches on the **7-variant** `ffi::TunnelStatus` [FFI §3.2] via the Dart-facing state diagram [FFI §3.4]. This is the spine the flows in §4–§6 map onto; it is reproduced here as the canonical reference (owned by [FFI §3.4], consumed here):



The UI mapping of each state (fill colour / icon / label / motion) is owned by the Home render matrix [SC §5] and the ConnectButton state table [CL §2]; the **flows below reference state names**, not colours, so they stay valid regardless of theme.

3. Flow 1 — first-run onboarding → first connect

Goal. A brand-new install reaches its first Connected{...} with the minimum decisions, never dead-ending, honouring the OS-mediated VPN-consent step.

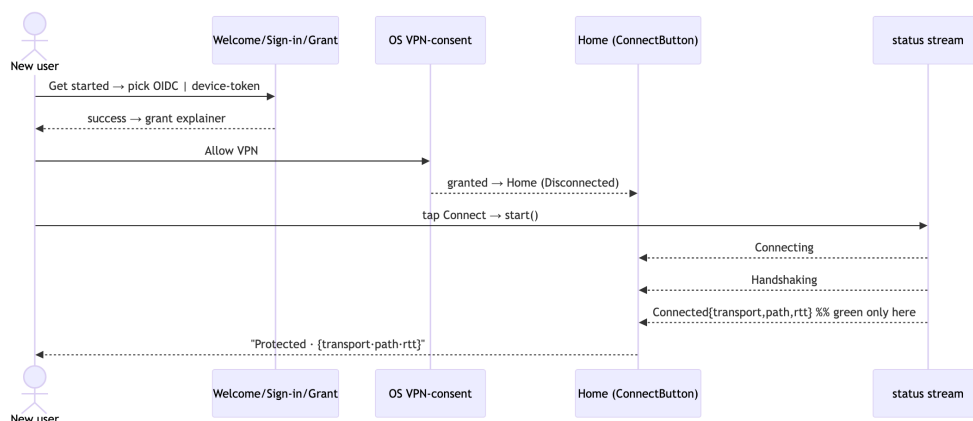
Entry. App launch with **no persisted device token / session** [04_CLIENT §8] → the launch gate routes to /welcome [SC §2]. (A present token skips straight to Home — Flow 2.)

Step-by-step.

1. **Welcome (/welcome, SC §4.1).** Static value-promise; one decision: **Get started** (→ sign-in) or "I already have an account / device". No async.
2. **Sign-in (/welcome/sign-in, SC §4.2).** The user picks **one of two** enrolment paths the spine supports [04_CLIENT §6]:
 - **OIDC** (managed identity / SSO) — opens the issuer in an OidcWebView / system browser; success returns a code exchanged for a session.
 - **Anonymous device token** (self-hostable, privacy-max) — generates + registers a per-device token against the configured server (server URL entered here on first run). No email, no personal identity. While the chosen path runs, the **other card disables** (no double-enrol); on failure an InlineError gives a typed, human reason (unreachable

server, bad URL, issuer rejected, token bound elsewhere) and returns to idle for retry — never a silent fail (§11.4.6).

3. **VPN permission grant (/welcome/grant, SC §4.3).** A pre-consent explainer ("Helix needs permission to create a VPN tunnel; we never see your traffic") then **Allow VPN** triggers the **OS-owned** consent dialog the TunnelPlatform shim requires (NEPacketTunnelProvider saveToPreferences / VpnService.prepare() / privileged-service install, per platform [SC §4.3]).
 - *granted* → brief success tick → Home.
 - *denied* → a **calm re-explain** + "Try again" + a "How to enable in Settings" deep-link; if the OS won't re-show the dialog (denied-permanently), the Settings deep-link is the path. Never a dead end.
4. **Home, Disconnected (/ , SC §5).** First post-onboarding frame. The ConnectButton renders Disconnected (the safe default — never a fake "connected" first frame) with chip "Tap to connect".
5. **First connect.** The user taps the ConnectButton → connectController.toggle(Disconnected) calls start() via the shim [04_CLIENT §8.2]. The status stream now drives the UI through Connecting → Handshaking → Connected{...} (Flow 2). The first Connected{...} event paints green — and **only** that event does (§11.4.107).



a11y. Each onboarding step is its own route (sane OS back; partial enrolment resumes at the right step); the actionable button is focused per step; success / denied results are announced via a polite live-region; the headline is an h1 heading node [SC §4]. **Resume:** a half-finished enrolment (token saved, consent not granted) resumes at step 3, not step 1.

4. Flow 2 — connect → established → graceful disconnect

Goal. The everyday happy path: connect, observe the established connection honestly (the §11.4.107 loading-vs-connected distinction), then disconnect cleanly.

The state transitions this flow maps to UI (the spine, §2):

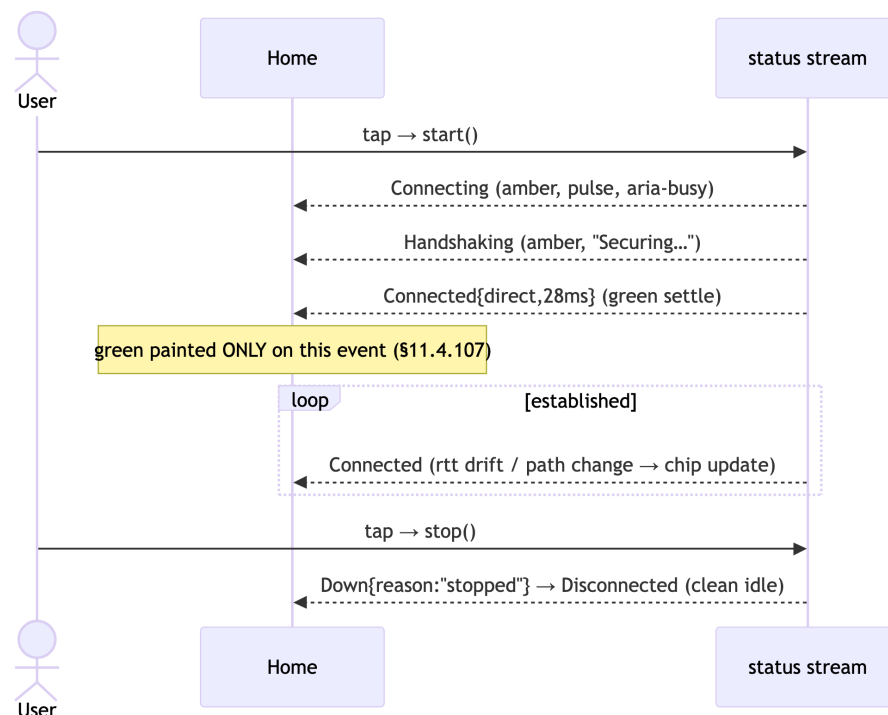
Transition	UI on Home [SC §5]	Why it is honest
Disconnected → Connecting	amber fill, spinner-ring, "Connecting..." , chip "Contacting exit...", connectPulse	animated + "..."-suffixed + non-committal chip ⇒ not "connected"
Connecting → Handshaking	amber (= Connecting), key-exchange glyph, "Securing..." , chip "Handshaking..."	still amber, still busy (aria-busy)
Handshaking → Connected{direct}	green fill, shield-check, "Protected" , chip transport·direct·{rtt}ms, one-shot settle	green painted only on the real Connected{...} event
Handshaking → Connected{relay}	teal-green, shield-check-relay, "Protected · relayed" , chip transport·relay·{rtt}ms	relay path surfaced honestly (not hidden as "direct")
Connected → Connected (rtt drift / path change)	chip updates in place; fill steady	re-emit coalesced [FFI §4.4]; no flicker
Connected → Disconnected (user stop)	grey, power-off glyph, "Not connected"	clean idle, distinct from a drop

Step-by-step.

1. **Tap ConnectButton** from Disconnected → start() [04_CLIENT §8.2].
2. **Connecting** — amber + connectPulse; the chip shows the in-flight phase; the button announces "Connecting" with aria-busy=true [CL §2].
3. **Handshaking** — "Securing..."; still amber, still busy. The user cannot mistake this for connected: colour (amber) + icon (spinner/key) + text ("..."-suffixed) + the non-committal chip are **four independent signals** that say "not yet" (§11.4.107, §13 colour-independence).
4. **Connected{...}** — green/teal **settle** (a one-shot, non-looping motion, §12); the chip now shows the live transport · path · rtt; the button

announces "Connected, direct path, 28 milliseconds" via a polite live-region. This is the **only** state painted as protected, and it is reached **only** from a real FFI Connected{...} event — never from the tap, a timeout, or an optimistic guess.

5. **Established steady state.** rtt drift / path change re-emit Connected, updating the chip without re-animating the fill (coalesced [FFI §4.4]). The shields row + exit summary reflect the live session.
6. **Graceful disconnect.** The user taps the ConnectButton (now in Connected) → toggle(Connected) calls stop(). The orchestrator emits Down{reason:"stopped"}, which the projector maps to **Disconnected** [FFI §3.3] (a *user* stop is clean idle, **not** a Down drop). The button cross-fades grey → "Not connected"; no Danger, no banner.



a11y. State changes fire polite live-region announcements; Connected adds the composed transport·path·rtt to the accessible name; the loading states set aria-busy; the disconnect tap is announced as a protection-reducing action only when a shield (kill-switch) would be lowered (it isn't here — a clean stop is neutral). **Connector parity:** the Connector Status home rides the same spine (its hero answers "am I up and serving my subnets?" [SN §0]) but with no user "connect" tap — an appliance is connected by config, so its home shows the live state read-only and surfaces re-enroll only on revoke (Flow 7).

5. Flow 3 — reconnect / drop recovery

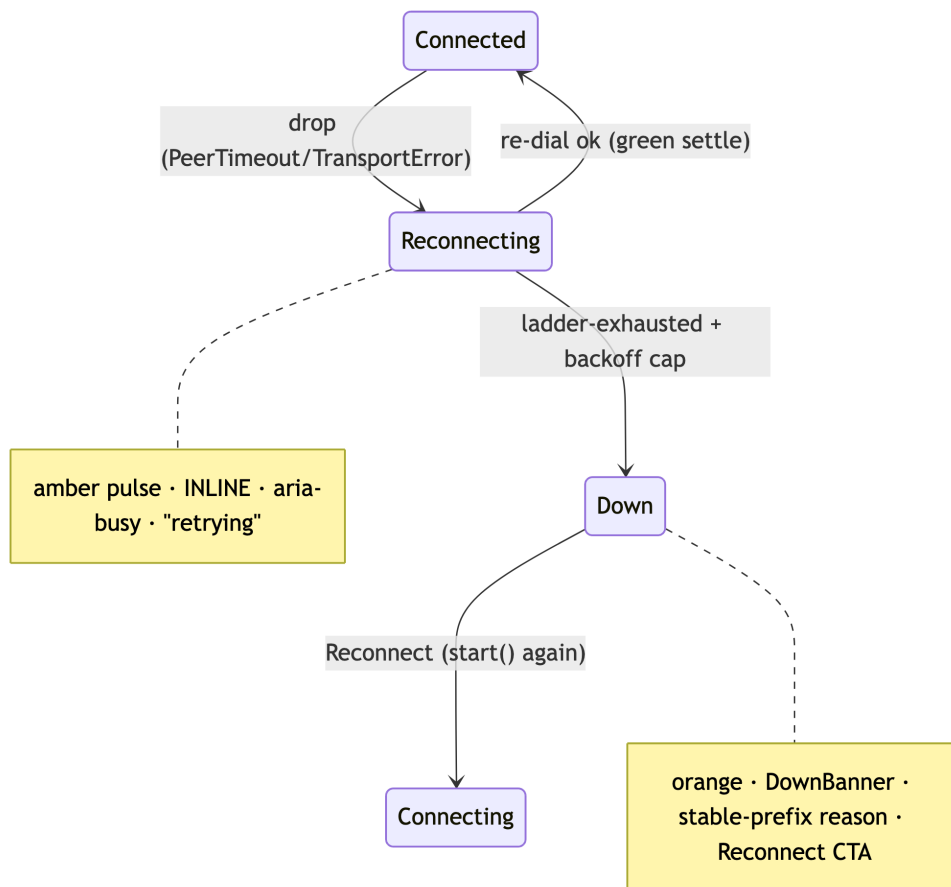
Goal. An *unexpected* link loss recovers automatically without alarming the user as if they were exposed — distinct from both a clean stop (Flow 2) and a Danger exposure (Flow 4).

The distinction (load-bearing, §11.4.6). Three different "not fully connected" situations get three different visual languages so the user is never misled:

Situation	FFI state	Home language	Severity
User stopped	Disconnected	grey, "Not connected"	neutral
Link dropped, retrying	Reconnecting	amber pulse , "Reconnecting...", chip "Lost link, retrying..."	transient (no exposure)
Retry budget exhausted	Down{reason}	orange + inline DownBanner on Home, "Disconnected — dropped", reason (stable-prefix)	failed (no exposure)
Traffic may be escaping	Danger{kind}	red full surface (Flow 4)	exposure

Step-by-step.

1. From Connected, the carrier drops (PeerTimeout / TransportError) → the orchestrator's re-dial driver begins; the FFI emits **Reconnecting**.
2. **Reconnecting** — amber **pulsing** fill, rotating glyph, "Reconnecting...", chip "Lost link, retrying..."; aria-busy=true. This is an **inline** state on Home — *not* a full red surface — because a retrying drop is not an exposure.
3. **Recovery path A — Reconnecting → Connected**. Re-dial succeeds → green settle; the chip returns to live transport · path · rtt. A polite announcement: "Reconnected, direct path, 31 milliseconds".
4. **Recovery path B — Reconnecting → Down{reason}**. The re-dial ladder is exhausted + the backoff cap is hit → **Down{reason}**: an orange button + the inline **DownBanner** on Home (z below Danger, never full-screen-red), label "Disconnected — dropped", the reason shown by **stable prefix** (ladder-exhausted / host-fatal) not parsed prose (§11.4.6). The banner offers **Reconnect** (→ start() again, back to Connecting).
5. **The kill-switch interaction**. If the kill-switch shield is **on**, traffic is blocked while Reconnecting/Down (the protection is *working*) — the shields row shows kill-switch active to reassure; this is coordinated by the screen, not an overlap [CL §5]. (If kill-switch is *off*, a drop may expose traffic → that path is a Danger{kind:"leak"}, Flow 4.)



a11y. Reconnecting announces politely (not assertively — it is not an emergency); Down announces the typed reason and moves focus to the DownBanner's Reconnect action. The pulse degrades to a **static** amber ring under reduce-motion (§12) — it never strobes (§11.4.107 no-flash).

6. Flow 4 — leak / kill-switch-tripped Danger (intent-override)

Goal. Make "you may be exposed" impossible to miss and impossible to suppress, and give exactly one safe action — without strobing or panicking.

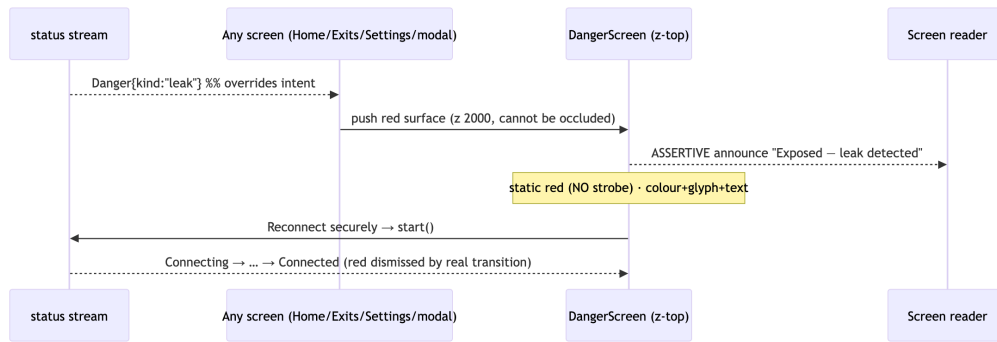
The override rule. A `Danger{kind}` from the stream **paints red regardless of where the user is** [FFI §3.3, §3.2]: the projector returns `Danger` *before matching any other state* ("Danger overrides everything"). The UI honours this by **pushing the Danger surface over the entire stack** — over Home, over Exits, over Settings, even over a modal or a toast (the `DangerBanner` is `z-top`, `z.semantic.danger 2000`, and can never be occluded [SC §0]). This is **D-UX-2**: the Danger surface is the only input-overriding navigation event in the product.

The two kinds.

Danger.kind	Source	Surface [SC §11]	Safe action
"leak"	shields_tripped = "leak" (DNS/IPv6/route escape) [FFI §3.3]	/danger/leak — full red DangerScreen + LeakDetail (what leaked, plain language)	Reconnect securely
"killswitch_tripped"	auth-failed (revoked) → kill-switch closed [FFI §3.3]	/danger/ killswitch — full red + KillSwitchDetail	resolve (re-sign-in / re-enroll)

Step-by-step.

1. The core detects an exposure (a shield trips, or auth fails and the kill-switch closes) → the FFI emits **Danger{kind}**.
2. **Immediate, anywhere.** Wherever the user is, the **DangerBanner** appears z-top (red, "EXPOSED — leak detected" / "Kill-switch engaged") and the full DangerScreen is pushed. An **assertive** screen-reader announcement interrupts (this is the one place an assertive announcement is correct, §13) because the user must know *now* [CL §2].
3. **No strobe.** The red is a **static** fill — there is **no flashing** (§11.4.107 no-flash; a strobing exposure warning would be both an ally hazard and a panic trigger). Colour + the alert glyph + the "Exposed" text are three independent signals so colour-blind and screen-reader users get it too (§13).
4. **One safe action.**
 - *leak* → **Reconnect securely** re-establishes the tunnel (→ Connecting); a secondary "See what happened" expands the LeakDetail.
 - *killswitch_tripped* (*auth revoked*) → resolve by re-sign-in / re-enroll (→ /welcome/sign-in, Flow 1 step 2); the kill-switch *stays closed* (blocking traffic) until resolved — the block is the protection working.
5. **Exit.** Danger → Connecting (reconnect / re-enroll) or Danger → Disconnected (user stop). The red surface is dismissed only by an actual state transition off Danger — never by a back-gesture that would merely hide an active exposure.



a11y. Assertive announcement (the only one in the product); focus moves to the single safe action; the red surface traps focus until a real state transition dismisses it; the static red clears the §11.4.107 no-flash rule.

Connector note: an appliance with kill-switch *off by default* [SN §0 principle 4] does not enter Danger{leak} on a drop the way a Client does — its exposure model is different (it serves a LAN); the Connector surfaces a drop as a health-degraded state, and a revoke as a re-enroll-required state, not a red consumer-exposure surface. This is **D-UX-3**.

7. Flow 5 — exit switch + multi-hop build

Goal. Change where traffic exits (single exit) or compose an ordered multi-hop chain — applied live, with honest validation, without dropping protection.

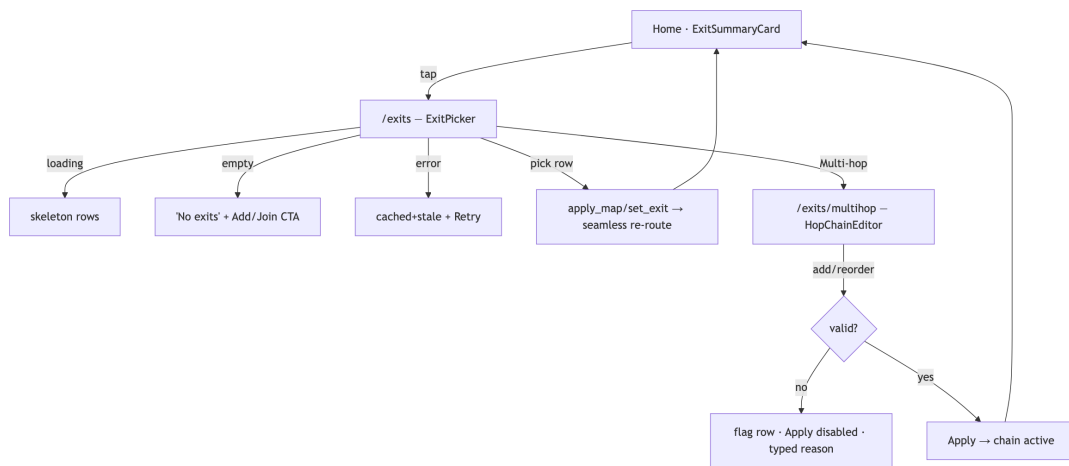
Sub-flow A — single exit switch.

1. From Home, tap the **ExitSummaryCard** → /exits (SC §6).
2. The ExitPicker list loads (core.exits()): **loading** = skeleton rows; **empty** = "No exits available" + a CTA (self-host with none configured) — never a blank screen; **error** = "Couldn't load exits" + retry + the last cached list shown stale-labelled [SC §6, §11.4.6]; **populated** = RTT-sorted ascending, favourites pinned.
3. The user filters (SearchField), optionally stars a favourite, taps a row.
4. **Apply live.** Selecting an exit while Connected triggers a **seamless re-route** (apply_map / set_exit [FFI §2]) — the orchestrator switches the active path without a full drop where possible; Home's ExitSummaryCard updates and the chip re-emits the new transport·path·rtt. The user returns to Home with the new exit active.

Sub-flow B — multi-hop build (SC §7, CL §6).

1. From Exits, tap "**Build a multi-hop chain**" → /exits/multihop.
2. The HopChainEditor opens with **one entry hop** (= the current exit). The user **adds** a second hop (an ExitPicker sub-sheet), then more (entry → middle → ... → exit).

3. **Live validation.** As hops are added/reordered, the summed estimated latency (Σ hop RTTs) recomputes in the sticky ChainSummaryBar. An **invalid** chain (duplicate hop, a non-relaying exit, overlapping-CIDR if a network hop) flags the offending row (feedback.error) and **disables Apply** with a typed reason — never silently accepted (§11.4.6).
4. **Reorder is keyboard/remote-operable**, not drag-only: focus a hop, Space to lift, arrow to move, Space to drop, with a live-region announcing "Hop 2 moved to position 1" (WAI APG reorder pattern [CL §6]).
5. **Apply** (enabled only when valid) calls apply_map/set_exit [FFI §2]; the chain becomes the active route; Home reflects it.



a11y. Each exit row announces name + jurisdiction + RTT + selected + favourite (RTT colour is paired with the numeric value — never colour-only [SC §6]); the favourite star is a nested button with its own 44 px target; each hop announces its ordinal + role ("Hop 1 of 3, entry, Sweden"); the chain summary announces total latency + validity.

8. Flow 6 — Console: enroll a device → author a policy → see it converge

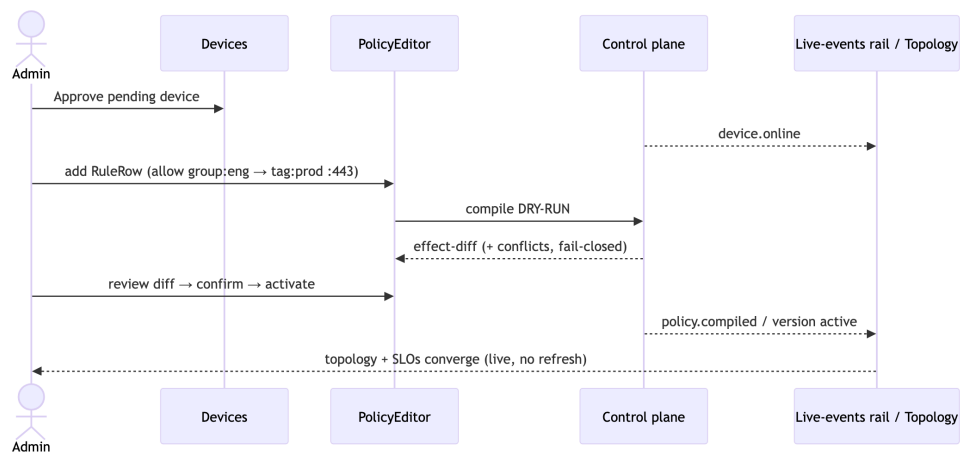
Goal. An admin admits a new device, writes an access rule for it, and watches the change propagate — fail-closed, preview-before-apply, live-not-poll [SK §0].

Step-by-step.

1. **Enroll / approve a device.** A new device appears in Devices (/devices, SK §5) as **pending-approval** (a PeerCard with primary **Approve** / **Deny** [CL §7]). The admin Approves → the control plane admits it; the live-events rail shows device.online [SK §1]. (Token minting for the device

itself is the Console operator action in [CP §9.2]; the device-side enroll is Flow 7 / Flow 1.)

2. **Author a policy (/policy, SK §7).** The admin adds a RuleRow in the PolicyEditor: an **AllowDenyToggle** (allow/deny), a **TargetPicker** source (e.g. group:eng), a destination (tag:prod), a port (:443) [CL §8]. Order matters (first-match ACL) — rows are keyboard-reorderable.
3. **Compile dry-run first (fail-closed).** The editor compiles the spec **dry-run** and shows an **effect-diff** before anything activates [POL §4/§5, SK §0]: what access this rule grants/revokes, and any **conflict** (a rule shadowed by an earlier one → amber conflictMark + a tooltip naming the shadowing rule [CL §8]). A compile error that names an un-advertised host CIDR **links to IPAM** (S6) — cross-links follow the data [SK §2].
4. **Activate.** Only after the admin reviews the diff and confirms (§11.4.66 confirmation for a high-stakes change) does the policy version activate; the live-events rail shows policy.compiled / version-activated.
5. **See it converge.** The admin watches convergence live: the topology graph (/topology, SK §8) and telemetry (/health) fold the WS/SSE stream so edges / device state update **without a manual refresh** [SK §0 principle 3]; the convergence/event-lag SLOs show the change landing. The **stream-health chip** is the §11.4.6-honest signal that "live" is real — amber when reconnecting, red
 - a "may be stale" notice when down, rather than lying [SK §1].



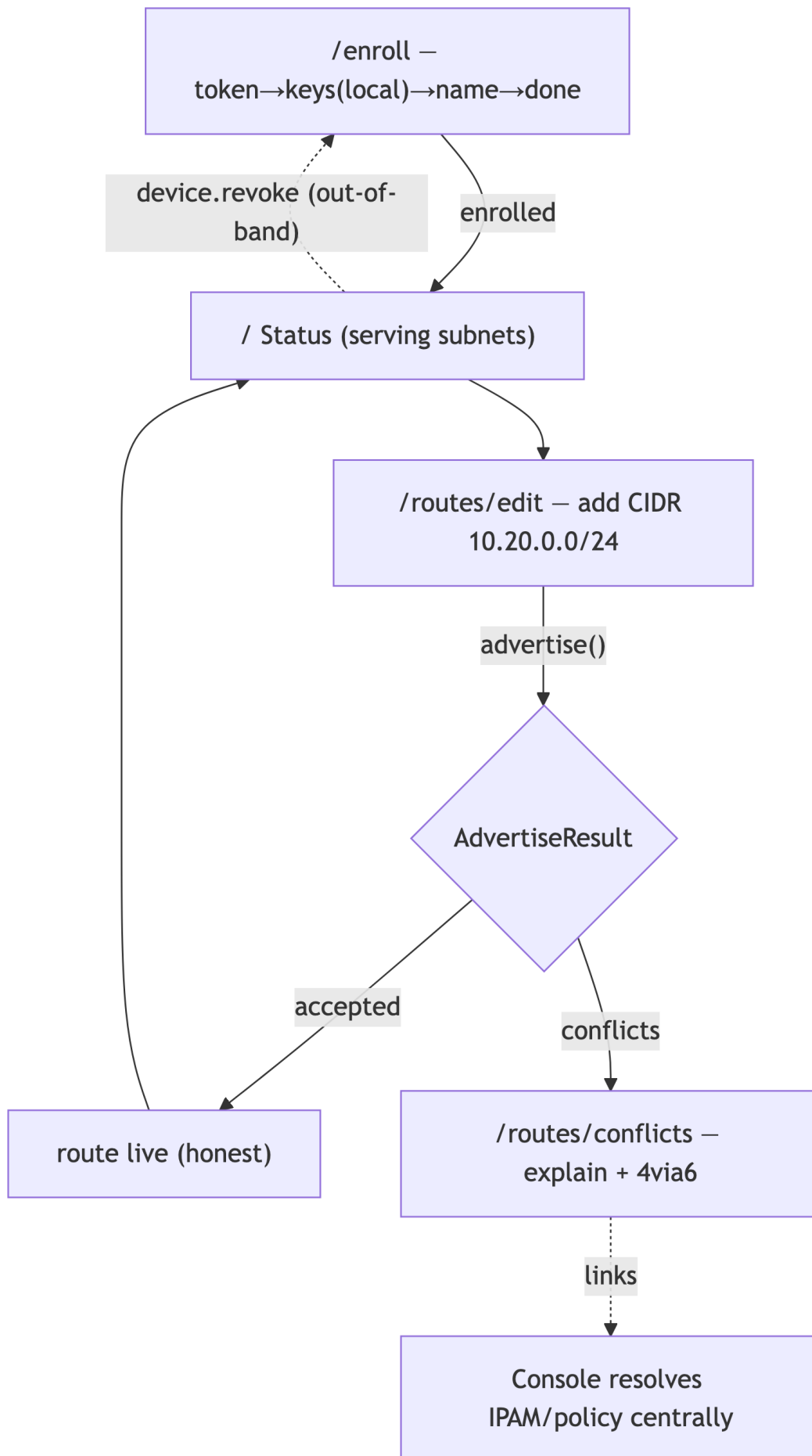
a11y. The policy is an ordered list of RuleRow listitems (order = precedence, announced "rule 3 of 8" [CL §8]); the allow/deny verb is colour + glyph (check / block), never colour alone; the effect-diff is a reviewable region; destructive control actions (revoke, deactivate) carry confirm dialogs [SK §0]. The topology graph ships a **dual representation** — the visual canvas plus an always-available navigable treegrid of the same nodes+edges — so a screen-reader / keyboard admin has a non-visual path (§11.4.117, [CL §9]).

9. Flow 7 — Connector: enroll → advertise a subnet → resolve a CIDR conflict

Goal. Stand up an appliance: turn an enroll token into an identity, declare the LAN subnet it exposes, and handle an overlapping-CIDR conflict honestly [SN §0].

Step-by-step.

1. **Enroll** (`/enroll`, SN §3). A first-run stepper (token → keys → name → done). The operator pastes the single-use **enroll token** (minted in the Console [CP §9.2]) or scans its QR. The device **generates its WG keypair locally — the private key never leaves it** (C6 [CP §9.2]); only the 32-byte public key registers. Steps show honest progress; on success → the Status home.
2. **Status home** (`/`, SN §4). The connector proves it is up and **serving its subnets** — the home is a pure function of the `statusStream` [FFI §3.2], never "up" on intent alone [SN §0 principle 1]. (Kill-switch is **Off by default** on an appliance — turning it on would sever the LAN it serves; that is a deliberate, explained choice [SN §0 principle 4, §7].)
3. **Advertise a subnet** (`/routes` → `/routes/edit`, SN §5). The operator adds a CIDR (e.g. `10.20.0.0/24`) and enables it. The core `advertise()` call returns `AdvertiseResult { accepted, conflicts }` [FFI §3.1]; the UI shows **accepted** vs **conflicting** CIDRs honestly and **never claims a conflicting prefix is live** [SN §0 principle 3].
4. **Resolve a conflict** (`/routes/conflicts`, SN §6). If the result carries a conflict (another site already advertises an overlapping CIDR), the conflict view **explains** it (with the `4via6 / overlapping-prefix` explanation [REG §6.2, IPAM §6]) and **links to the Console**, which resolves IPAM/policy **centrally** — the Connector view explains + guides, it does not resolve centrally itself [SN §0, SN §2]. The operator either changes the local CIDR or coordinates the central fix.
5. **Out-of-band revoke**. A `device.revoke` from the Console [CP §9.3] drops the Connector back to an **enroll-required** state (→ `/enroll`), surfaced honestly on the Status home, not as a silent failure.



a11y. The enroll stepper announces "step N of M" + the current step's purpose; the keygen step states the private key never leaves the device (privacy-honest); each route row announces its CIDR + accepted/conflict state (never colour-only); the conflict view's explanation is body text + a link, focus-ordered. The appliance form factor is compact-first, so the bottom-nav (Status · Routes · Settings) is the default affordance [SN §1].

10. Universal interaction patterns

These patterns are shared by all three apps; a flow above composes them.

10.1 Gestures (touch)

Gesture	Meaning	Where	Reduce-/no-touch fallback
tap	activate the primary action	everywhere	keyboard Enter/Space; D-pad OK
long-press	reveal context actions (§10.4)	list rows, ConnectButton (transport-override sheet UNVERIFIED)	keyboard menu key / a visible ... button
swipe (horizontal)	onboarding carousel; reveal row actions (favourite/remove) where used	Welcome pager [SC §4.1]; list rows	arrow keys; visible inline action buttons
drag	reorder (hops, policy rules) — sugar only	multi-hop, PolicyEditor	keyboard lift/move/drop is the canonical path [CL §6] a visible
pull-to-refresh	re-probe live data (RTT, devices)	Exits list, Console tables	Refresh button; live stream where present (§10.5)
pinch / pan	zoom the topology graph	Console topology	zoom/fit controls + keyboard pan [CL §9]

Every gesture has a non-gesture equivalent (§13) — the canonical path is never gesture-only. Gesture thresholds are UNVERIFIED (owned by [platform-adaptation.md]).

10.2 Keyboard navigation (desktop / web-less native)

- **Tab order** follows reading order (top→bottom, leading→trailing); modals **trap** focus; Esc closes dismissibles and **restores focus** to the trigger (e.g. ExitPicker trigger [CL §4.3]); tab order never jumps [CL §1.3].
- **Activation:** Enter / Space on buttons + the ConnectButton toggle; arrow keys traverse lists/rails/segmented controls; Home/End jump list ends; PageUp/Down page long lists [CL §4.2].
- **Reorder:** focus → Space lift → arrows move → Space drop, with live-region feedback (multi-hop, policy rules) [CL §6].
- **Console command palette:** ⌘K / Ctrl-K jumps to any screen / entity / action; keyboard-first, but every action it lists is also mouse-reachable [SK §1].
- **Focus is never colour-only:** a 2 px border . focus ring, offset from the container, visible against any state fill (≥3.0 non-text contrast both themes [COLOR §4.5]) [CL §1.2].

10.3 TV-leanback D-pad / remote

- **Default focus** is declared per screen (Client Home → ConnectButton; Welcome → "Get started"; Appearance → the current theme choice) so the remote lands on the right control [SC §4, §5, §10.a].
- **Traversal:** D-pad up/down/left/right moves focus along the per-screen declared order; OK/center activates; Back pops. No hover-only affordances; no gesture-only paths.
- **Enlarged targets + ring:** focus targets are enlarged to ~64 px hit area with an always-visible focus ring (border . focus); the rail is reachable by D-pad (no bottom-nav on TV) [SC §0, §3].
- The exact per-screen D-pad focus-order matrices are UNVERIFIED — pinned by [platform-adaptation.md] + the leanback golden-screenshot suite; this doc and the screens-* docs state the focus **intent** per screen [SC §0].

10.4 Long-press / context actions

Long-press (touch) / menu-key (keyboard) / long-OK (remote) reveals **secondary** actions that are never the only path to a capability:

- **Client:** long-press the ConnectButton → a quick transport-override sheet (auto / specific) [SC §5] (UNVERIFIED — the override UX is pinned

by [CL §2]); long-press an exit row → favourite/details (also reachable by the visible star + row tap).

- **Console:** a PeerCard / AuditLogRow exposes its actions via a visible ---menu button **and** the high-priority actions inline (Approve / Revoke), so the menu is an accelerator, not a requirement [CL §7, §10].

Every long-press target also exposes a **visible** affordance (a --- button, a star, an inline action) so the capability is discoverable without knowing the gesture (§13).

10.5 Pull-to-refresh & live-vs-pull

HelixVPN distinguishes **live** surfaces (folding a stream) from **pull** surfaces (on-demand re-probe). This is **D-UX-4**:

- **Live surfaces** (Client Home status, Console device list / topology / audit / live-events rail, Connector Status) fold the FFI status stream or the control plane's WS/SSE /v1/stream and update **without a manual refresh** [SK §0, FFI §4]. A **stream-health chip** honestly signals stream state (green / amber reconnecting / red down + "may be stale" notice) — the live claim is never a lie [SK §1].
- **Pull surfaces** (Exit-list RTT re-probe, Console tables on a non-stream view) support **pull-to-refresh** + a visible Refresh button; a pull shows a determinate-where-possible progress, then the fresh (or cached+stale-labelled on failure) data [SC §6].

A surface is **never both lying-fresh and stale**: if a live stream is down, the data is shown with an explicit stale label, not silently frozen as fresh (§11.4.6).

11. The loading / empty / error / success state contract

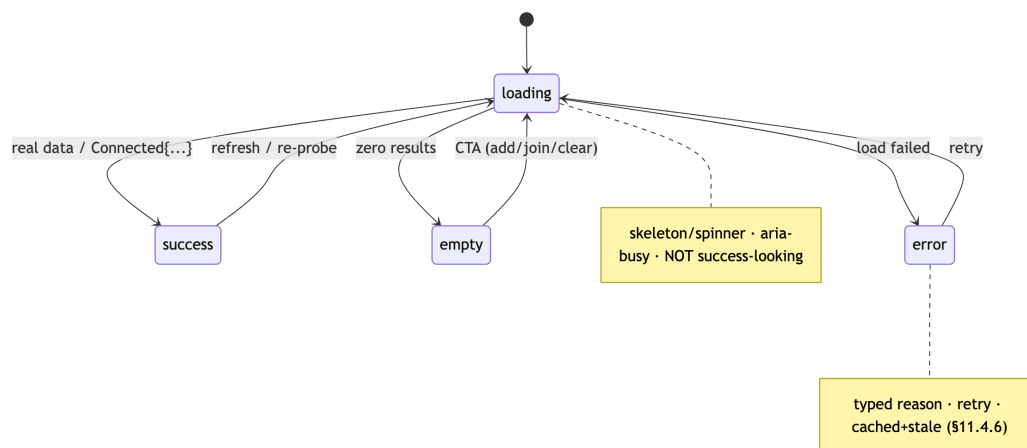
Every async surface in every app renders one of **four** states, each a reusable contract (composed from [CL §11.15] ProgressIndicator, [CL §11.16] EmptyState, [CL §11.17] Skeleton, [CL §11.8] Banner). This is **D-UX-5** — the single state grammar all three apps share.

State	What it MUST do	What it MUST NOT do
loading	show a skeleton (list/table) or an indeterminate spinner (action), aria-busy=true, distinct colour/motion from any success state	never look like the connected/success state (§11.4.107); never block forever with no timeout path

State	What it MUST do	What it MUST NOT do
empty	state the empty reason + offer a next action (CTA: "Add a server" / "Join a network" / "Clear filter")	never a blank screen; never imply data exists
error	a typed, human reason + retry + show the last cached value stale-labelled if available (§11.4.6)	never a silent fail; never a raw stack/error code as the only message; never fake-fresh cached data
success	render the real data; for the connection hero, paint the protected/connected language only on a real Connected{...} event	never paint success off a timer, a tap, or an optimistic guess (§11.4.107)

The §11.4.107 loading-vs-connected rule is the contract's keystone. On the connection hero (Client Home, Connector Status), "loading/connecting" and "connected" are kept distinct by **four independent signals** — colour (amber vs green/teal), icon (spinner/key vs shield-check), text ("..."-suffixed vs "Protected"), and the chip (non-committal vs live transport · path · rtt) — so the distinction survives colour-blindness, a muted screen, and a screen reader (§11.4.107, §13). A spinner is **never** proof of connection.

State transitions cross-fade fill **and** label together (motion.semantic.stateXfade, 180 ms [DT §6.4]) so no in-between frame is illegible (§12, [SC §0]).



12. Motion choreography

Motion is **functional, not decorative** — it communicates state change, spatial relationship, and busy-ness, and it always degrades to a static end-state under reduce-motion. All durations/curves are **tokens** [DT §6.4]; this doc names the *choreography*, the values live in tokens.

Moment	Motion token	Behaviour	Reduce-motion fallback
state cross-fade (any state→state on the hero)	<code>motion.semantic.stateXfade</code> (180 ms)	fill and label cross-fade together; no illegible mid-frame	instant swap to the end-state
connecting / reconnecting	<code>motion.semantic.connectPulse</code> (looping aura)	amber pulse aura while Connecting/ Handshaking/ Reconnecting	static amber ring (no loop, no strobe)
connected settle	one-shot settle	a single, non-looping "lock-in" on reaching <code>Connected{...}</code>	static green/teal end-state
press feedback	<code>motion.semantic.press</code> (≈100 ms)	a brief scale/overlay on tap	overlay only, no scale
danger	none	static red — no strobe, no flash (§11.4.107)	already static
skeleton shimmer	<code>comp.skeleton shimmer</code>	a low-contrast sweep while loading	static skeleton (<code>motion.semantic.reducedMotion</code>)
route/sheet transitions	<code>motion.semantic.* page/sheet</code>	enter/exit slide+fade	fade only / instant

Hard rules. (1) **Nothing strobos** — the Danger surface and every pulse are static or sub-flash-threshold (§11.4.107 no-flash; the connecting pulse is a slow aura, not a flash). (2) **Every looping/transition animation degrades** to its static end-state via `motion.semantic.reducedMotion` [DT §6.4, CL §1.3] when the OS reduce-motion setting is on. (3) **Motion never conveys state alone** — it reinforces a colour+icon+text signal, never replaces it (a reduce-motion user loses no information).

13. Accessibility (WCAG 2.1 AA)

HelixVPN targets **WCAG 2.1 Level AA** across all three apps and eight platforms. This is **D-UX-6**. The baseline (every component) is owned by [CL §1.3]; this section owns the **flow-level** a11y guarantees.

Screen-reader flow.

- Every flow above is fully operable by screen reader (VoiceOver / TalkBack / Narrator / Orca / the HarmonyOS+Aurora a11y trees [CL §1.3]); the canonical path is never gesture- or hover-only.
- **Connection state is announced** on every change: the ConnectButton announces a **stateful** name ("Connecting" / "Connected, direct path, 28 milliseconds" / "Not protected, leak detected") via a **polite** live-region — except Danger, which fires an **assertive** announcement (the one place interruption is correct, §6) [CL §2].
- Loading states set `aria-busy`; the StatusChip is a polite status live-region [CL §3]; error states move focus to the error/retry control.

Focus order. Logical (reading order); modals + the Danger surface **trap** focus; Esc closes dismissibles and restores focus to the trigger; the multi-hop and policy reorders are keyboard lift/move/drop with live-region feedback [CL §1.3, §6].

Contrast. Text vs background ≥ 4.5 AA-normal (or 3.0 AA-large for ≥ 24 px); non-text UI parts ≥ 3.0 ; every state fill is proven against its white/dark label in both themes [COLOR §4]. The focus ring clears the 3.0 non-text floor in both themes [COLOR §4.5].

Reduced motion. Every looping/transition animation degrades to its static end-state (§12); nothing strobes (§11.4.107 no-flash).

Hit targets. Every interactive target is $\geq 44 \times 44$ px (touch floor); a visually-smaller control keeps a 44 px transparent tap area centred on the glyph; TV-leanback enlarges to ~ 64 px (a hard gate CM-hit-target-44 [CL §1.3, visual-regression-and-qa.md]).

Colour-independent state (the load-bearing a11y rule for a VPN product). State is **never** conveyed by colour alone — every coloured state also carries an **icon and text**: connecting (amber + spinner + "Connecting..."), connected (green

- shield-check + "Protected"), reconnecting (amber-pulse + rotate + "Reconnecting..."), down (orange + alert + "dropped"), danger (red + alert + "Exposed"). A colour-blind user reads "Protected" vs "Exposed" from the **glyph + label**, not the green/red [COLOR §0, CL §2, §11.4.107]. RTT, presence dots, allow/deny verbs, and audit severities are all icon/text-reinforced too.

Graph fallback. The Console topology ships a **dual representation** — the visual canvas **plus** an always-available navigable treegrid of the same nodes+edges — so a non-visual user has a structured path (§11.4.117, [CL §9]).

The §11.4.162 covenant. Every screen ships light AND dark, no element overlaps another, no element overlays a label, and visual-regression goldens (per theme, per size, per app) OCR-/overlap-check every surface [CL §1.5, SC §0].

14. Internationalization, RTL & localization

HelixVPN is multi-region (the spine targets self-host + managed worldwide), so every flow is **i18n-ready by construction**. This is **D-UX-7**.

String externalization. No user-visible string is hard-coded; all copy is in ARB/.arb-class resource bundles resolved at runtime (UNVERIFIED exact l10n toolchain — owned by [platform-adaptation.md]). Status labels ("Connecting...", "Protected", "Exposed"), error reasons, and CTAs are localized; the **stable reason prefixes** the FFI carries (ladder-exhausted, auth-failed, killswitch_tripped [FFI §3.3]) are **keys**, mapped to localized human strings in the UI layer — never the raw key shown to a user, never the localized string parsed back (§11.4.6).

RTL (right-to-left). Every layout mirrors for Arabic / Hebrew / Persian: leading/trailing swap (the favourite star, chevrons, the → policy arrow, the multi-hop chain direction all mirror); the bottom-nav / rail order mirrors; text aligns to the start edge. Directional glyphs (chevron, arrow, reorder) use **start/end** semantics, not fixed left/right, so they flip correctly. The multi-hop chain renders entry → ... → exit start-to-end, mirroring in RTL so the *meaning* (entry first) is preserved [CL §6].

Text expansion. Layouts tolerate ~+30–40 % string growth (German / Finnish): the no-overlap covenant (§11.4.162) means a label grows within its own box and ellipsises + tooltips rather than colliding with a neighbour [CL §1.5]; the ConnectButton label wraps (never truncates) under large accessibility text sizes

- long localizations, never overlaying the icon or sub-label [SC §5].

Numerals, dates, units. RTT ("28 ms"), latency sums ("+71 ms"), timestamps (audit UTC), and counts use locale-aware formatting; the audit log keeps UTC for forensic precision but announces a locale-readable form [CL §10]. Country/flag labels in the exit list are localized country names, not only flags (the flag is decorative; the name carries meaning, §13).

Bi-di safety. Mixed-direction content (an LTR server URL or CIDR inside an RTL sentence) uses bidi isolation so a CIDR like `10.20.0.0/24` or a URL renders correctly within RTL copy.

15. Surfaced decisions & cross-doc contracts

ID	Decision	Status
D-UX-1	Bottom-nav (compact) / rail (medium+) are the primary nav affordances; drawer is secondary-overflow only (Console "More" sheet on compact).	Asserted (this doc)
D-UX-2	The Danger{kind} surface is the only input-overriding navigation event; it pushes z-top over everything, dismissed only by a real state transition off Danger.	Grounded [FFI §3.3, SC §11]
D-UX-3	The Connector's exposure model differs from the Client's (kill-switch Off by default; a drop is health-degraded, a revoke is re-enroll-required — not a red consumer-exposure surface).	Grounded [SN §0, CP §9.3]
D-UX-4	Live surfaces (status / device list / topology / audit) fold a stream + a stream-health chip; pull surfaces (RTT re-probe, non-stream tables) use pull-to-refresh; a surface is never lying-fresh while stale.	Asserted (this doc); grounded [SK §0, FFI §4]
D-UX-5	One loading / empty / error / success state grammar shared by all three apps; the §11.4.107 loading-vs-connected distinction is its keystone.	Asserted (this doc)
D-UX-6	WCAG 2.1 AA is the accessibility target across all apps/platforms; colour-independent state (icon+text) is the load-bearing rule for a VPN product.	Asserted (this doc)
D-UX-7	Every flow is i18n-ready by construction: externalized strings, full RTL mirroring (start/end semantics), +30–40 % expansion tolerance, locale-aware numerals/dates, bidi isolation.	Asserted (this doc)

Cross-doc contracts this doc depends on.

- The **7-variant ffi::TunnelStatus** state machine + Dart-facing state diagram [FFI §3.2, §3.4] — the spine §2–§6 ride; if the FFI enum changes, the flows here re-map.

- The **Home render matrix** [SC §5] + **ConnectButton state table** [CL §2] — the per-state colour/icon/label/motion this doc references by state name.
- The **connection-state palette** + contrast proofs [COLOR §3, §4] and **motion tokens** [DT §6.4] — every colour/motion reference resolves there.
- The three **screen inventories** + **nav maps** [SC §2, SK §2, SN §2] — the per-app IA §1 consumes.
- The **component a11y baseline** [CL §1.3] + **topology dual-representation** [CL §9] — the §13 flow-level a11y builds on.
- The §11.4.162 **no-overlap / light+dark / visual-regression** covenant [CL §1.5, SC §0] — enforced on every surface a flow renders.

UNVERIFIED surfaces in this doc (flagged inline, never fabricated): exact gesture thresholds + per-screen TV D-pad focus-order matrices ([platform-adaptation.md]); the ConnectButton transport-override long-press UX ([CL §2]); the l10n toolchain ([platform-adaptation.md]); the exact router config (deferred to a Volume-10 navigation refinement pass — see MASTER_INDEX; not yet authored).

Sources verified

- **HelixVPN sibling specs (same wave, on disk, read for consistency):**
 final/v10-design/screens-client.md (Client screen inventory, nav map, AdaptiveScaffold ladder, Home 7-state render matrix, Danger surfaces) · final/v10-design/screens-console.md (Console IA, command palette, live-events rail, policy fail-closed, stream-health chip) · final/v10-design/screens-connector.md (Connector near-linear IA, enroll, advertise, conflict, appliance defaults) · final/v10-design/component-library.md (ConnectButton/StatusChip state sets, reorder pattern, topology dual-representation, a11y baseline, no-overlap guarantee) · final/v10-design/color-system.md (connection-state palette, contrast proofs, focus-ring contrast) · final/v04-client/ffi-surface.md §3 (the 7-variant ffi::TunnelStatus, projector, §3.4 Dart-facing state diagram, §4 StreamSink delivery + current-truth guarantee). Verified 2026-06-25 against the files at the cited paths.
- **External interaction/accessibility precedent** (pattern grounding, not copied): W3C **WAI-ARIA Authoring Practices Guide** (listbox, dialog focus-trap + restoration, keyboard reorder, treegrid) — <https://www.w3.org/WAI/ARIA/apg/> · **WCAG 2.1** Level AA success criteria (1.4.3 contrast, 1.4.11 non-text contrast, 2.1.1 keyboard, 2.3.1 no-flash, 2.5.5 target size, 1.4.10 reflow, 1.3.4 orientation) — <https://www.w3.org/TR/WCAG21/> · **Material Design 3** adaptive navigation (bottom-nav / rail / drawer by window-size class) — <https://m3.material.io/foundations/layout/understanding-layout/overview> · **Apple HIG** — Designing for

tvOS (focus + parallax, remote interaction) — <https://developer.apple.com/design/human-interface-guidelines/designing-for-tvos> · **Android TV** leanback / D-pad focus guidance — <https://developer.android.com/training/tv/start/navigation> . Verified 2026-06-25 (latest-source check per §11.4.99; these are stable W3C/vendor references used for pattern grounding only — no HelixVPN behaviour is asserted from them).

- **Constitution anchors applied:** §11.4.6 (no-guessing — UNVERIFIED tags, no fabrication), §11.4.107 (loading ≠ connected — the keystone of §4, §11, §12), §11.4.117 (pixel-only graph → structured fallback, §13), §11.4.162 (OpenDesign: light+dark, no overlap / no label overlay, visual regression — §11, §13).